

COMP3710 · PATTERN ANALYSIS

Demonstration 2

Pattern Recognition

What each part asks of you, and what you should take away from it.

Week 1 of 4 · overview, Parts 1 to 3

Where this demo sits

Week	Date	Lecturer	Topics	Lectures	Pracs	Assessments
1	27/07/2026	Shakes	Patterns - Symmetry and Self Similarity	Intro	Fractals	
2	3/08/2026			Groups Workshop Group Theory		
3	10/08/2026			Fractals/Chaos Workshop Fractals/Chaos Theory		
4	17/08/2026		Traditional Pattern Recognition	Patt. Recog. Workshop Fourier/Conv Theory	Demo Code Due and Marked Off	
5	24/08/2026			Features Workshop PCA/RF Theory		
6	31/08/2026	Wei	Deep Learning 1 - Convolutional Neural Networks	ML/Classifiers Workshop	Pattern Recognition	
7	7/09/2026			NN Theory NN Workshop CNN Theory		
8	14/09/2026			Segmentation Workshop Autoencoder Theory VAE/Diffusion Workshop		Demo Code Due and Marked Off
9	21/09/2026			Visualisation Theory Symposium 1		
Break	28/09/2026			Break		
10	5/10/2026	Shakes	Deep Learning 2 - Transformers & LLMs	Transformers Theory	Report	
11	12/10/2026			Symposium 2 Multi-agent systems Theory		
12	19/10/2026			Report Workshop		
13	26/10/2026		State-of-the-art Models	VNN Theory VNNs Workshop	Demo Code Due and Marked Off	
			Course Review	Summary/Review Exam Prep		

Course Schedule

Course Schedule

The course schedule — Demonstration 2 covers Traditional Pattern Recognition

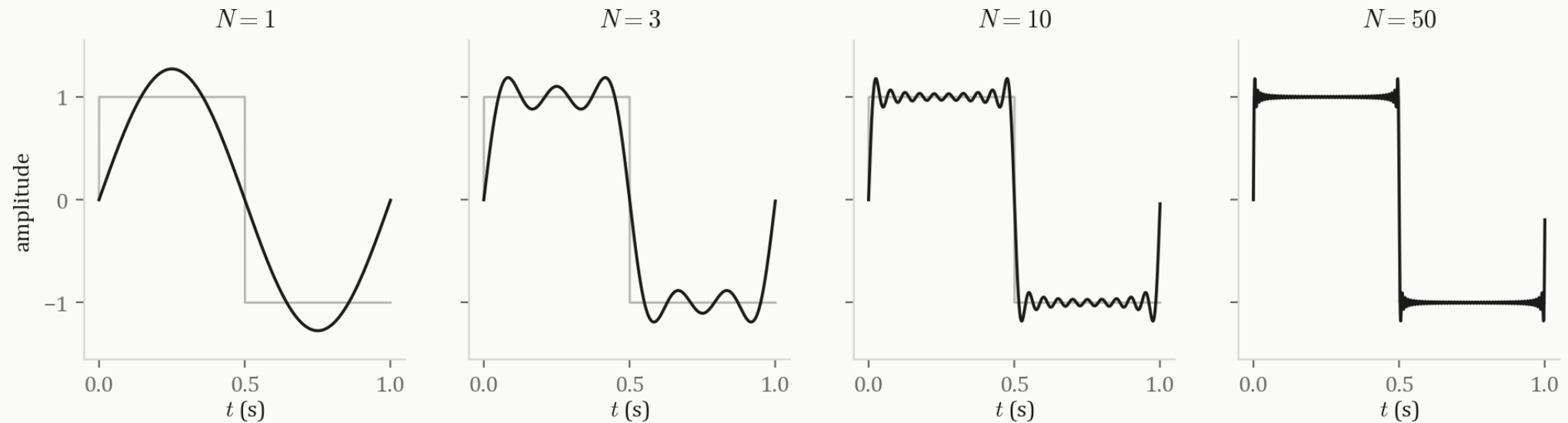
What Demo 2 covers

- The tasks demonstrate different PyTorch skills, using things taught in the lectures as examples.
- They build up to developing and training your own neural networks for the last part of Demo 2 — and for your project.

Q	SKILL	TASK
1	Revising PyTorch and GPU operations	Discrete Fourier Transform of an approximate square wave
2	Loading and visualising an image dataset; PCA; a random forest classifier	Classifying the Faces in the Wild dataset
3	Implementing and training CNNs	CNN models — a simple face classifier, then ResNet-18 on CIFAR-10
4	Combining the above to research, implement and train your own networks	Three architectures: VAE, UNet, GAN

Discrete Fourier Transform

A signal written two ways: as samples in time, and as sinusoids in frequency.



Fourier series with N terms, against the true square wave (grey)

Part 1 — tasks

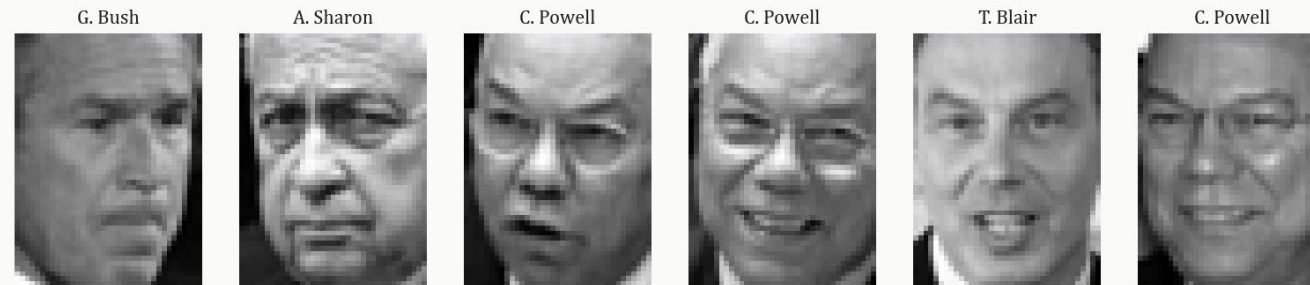
1. Start by building an approximate square wave using Fourier series approximations.
 - Play around with the number of components (harmonics) used in the approximation.
2. We give you example code for the DFT — you'll adapt and use it on your approximate square wave.
3. The DFT decomposes a function into its frequencies — compare the result to the harmonic functions used to approximate the square wave in the first place.
4. Then, practice porting the square wave and DFT implementations (originally in NumPy) to PyTorch, and run them on GPU.
 - Compare runtime between your GPU implementation and the original CPU version.

Part 1 — what you should learn

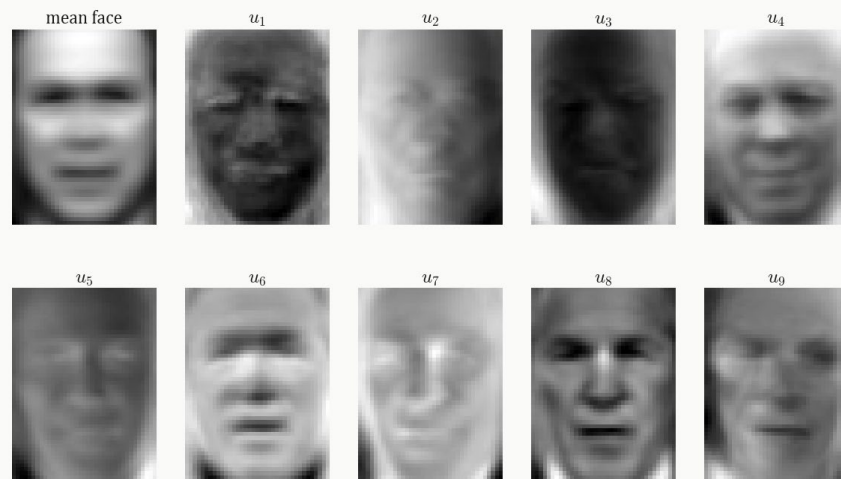
Take away from this part

- Fourier series and the Discrete Fourier Transform — how a signal decomposes into frequencies.
- Writing PyTorch code and running it on GPU.

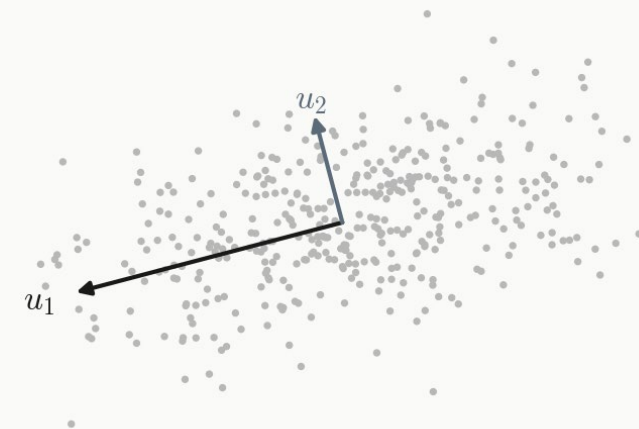
Eigenfaces



Labeled Faces in the Wild — example images



Eigenfaces: the mean face and the top directions



PCA in 2D: directions of greatest variance

Part 2 — tasks

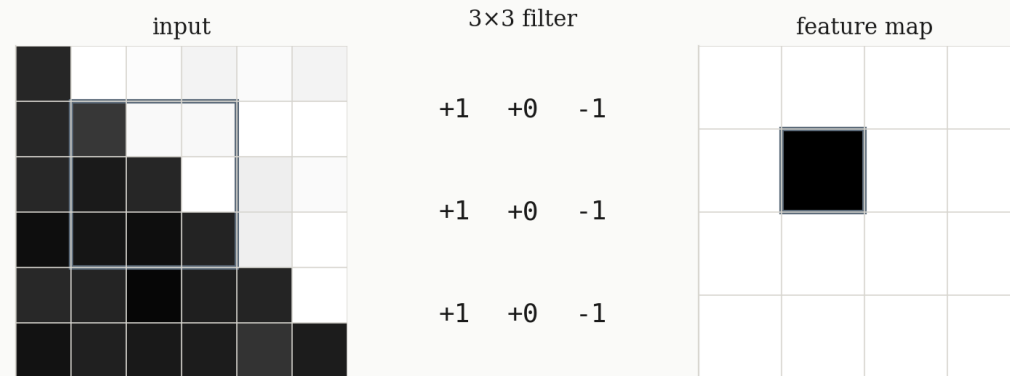
- Run the given code to decompose the faces into their principal directions (PCA / eigenfaces).
- Run the given code to classify the faces using a random forest.

Part 2 — what you should learn

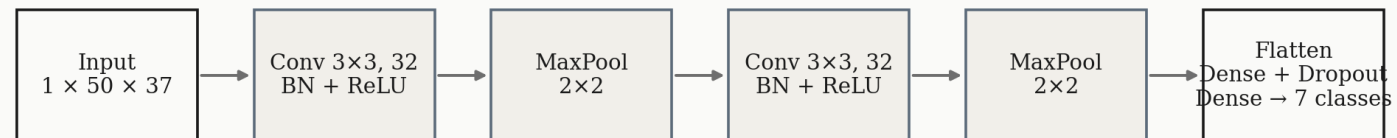
Take away from this part

- The PyTorch dataloading pattern used here — it can be reused directly in Q3.1, and again through the rest of Q3 and Q4.
- PCA — a useful tool for visualisation in lots of deep learning and computer vision work.

Convolutional Neural Networks



A filter convolved over the input produces one feature map



The CNN architecture you will implement for Part 3.1

Part 3 — tasks

1. Implement a very simple (3-layer) CNN to classify the faces from Part 2.
2. Implement a ResNet-18 CNN model for CIFAR-10 classification.
 - CIFAR-10: 32×32 images across 10 natural-image classes — see below.
 - Train it on Rangpur.



CIFAR-10 classes, one example image each

Part 3 — what you should learn

Take away from this part

- How to implement a CNN.
- How to train a deep learning model: optimiser, loss, normalisation, training loop.

Recognition

COMING SOON Content for this part is still being put together.